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Department Of Software Engineering

Cloud Computing Lab

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Section: 5-B

LAB NO 12

Lab 12 – Terraform Provisioners, Modules & Nginx Reverse Proxy/Load Balancer

Task 0 Lab Setup (Codespace & GH CLI)

- task0_codespace_create_and_list.png

```
PS C:\Users\saira> gh repo create CC_Saira_2023-BSE-057-Lab12 --public
✓ Created repository 23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12 on github.com
https://github.com/23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12
```

```
PS C:\Users\saira\Lab12> git init
Initialized empty Git repository in C:/Users/saira/Lab12/.git/
PS C:\Users\saira\Lab12> echo "# CC_Saira_2023-BSE-057-Lab12" > README.md
PS C:\Users\saira\Lab12> git add README.md
PS C:\Users\saira\Lab12> git commit -m "Initial commit"
[master (root-commit) 822882d] Initial commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 README.md
PS C:\Users\saira\Lab12> git branch -M main
PS C:\Users\saira\Lab12> git remote add origin https://github.com/23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12.git
PS C:\Users\saira\Lab12> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 277 bytes | 138.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\saira\Lab12> gh codespace create --repo 23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12 --branch main
✓ Codespaces usage for this repository is paid for by 23-22411-057-sudo
? Choose Machine Type: 2 cores, 8 GB RAM, 32 GB storage
vigilant-space-spork-q7579p9pwjq6cxj4j
PS C:\Users\saira\Lab12> gh codespace list
```

NAME	DISPLAY_NAME	REPOSITORY	BRANCH	STATE	CREATED_AT
studious-fishstick-r454rprpjx94f6p7	studious fishstick	23-22411-057-sudo/exam_prep	main*	Shutdown	about 28 days ago
vigilant-system-5g7g4x4xw74737p4w	vigilant system	23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab11	main*	Available	about 22 hours ago
vigilant-space-spork-q7579p9pwjq6cxj4j	vigilant space spork	23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12	main	Available	less than a minute ago

```
PS C:\Users\saira\Lab12> gh codespace ssh -c vigilant-space-spork-q7579p9pwjq6cxj4j
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab12 (main) $ |
```

- task0_codespace_ssh_connected.png

```

PS C:\Users\saira\Lab12> gh codespace ssh -c vigilant-space-spork-q7579p9pwjq6cxj4j
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab12 (main) $ |

```

Task 1 — Organize Terraform code into separate files

- task1_project_directory.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab12 (main) $ mkdir -p ~/Lab12
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab12 (main) $ cd ~/Lab12
@23-22411-057-sudo → ~/Lab12 $

```

- task1_files_created.png

```

@23-22411-057-sudo → ~/Lab12 $ touch main.tf variables.tf outputs.tf locals.tf terraform.tfvars entry-script.sh
@23-22411-057-sudo → ~/Lab12 $ ls
entry-script.sh  locals.tf  main.tf  outputs.tf  terraform.tfvars  variables.tf
@23-22411-057-sudo → ~/Lab12 $ |

```

- task1_variables_tf.png

```

GNU nano 7.2                                     variables.tf *
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "instance_type" {}
variable "public_key" {}
variable "private_key" {}
|

```

- task1_outputs_tf.png

```

GNU nano 7.2                                     outputs.tf *
output "aws_instance_public_ip" {
  value = aws_instance.myapp-server.public_ip
}
|

```

- task1_locals_tf.png

```

GNU nano 7.2                                     locals.tf *
locals {
  my_ip = "${chomp(data.http.my_ip.response_body)}/32"
}

data "http" "my_ip" {
  url = "https://icanhazip.com"
}

```

- task1_terraform_tfvars.png

```
GNU nano 7.2 terraform.tfvars *
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
```

- task1_main_tf.png

```
GNU nano 7.2 main.tf *
public_key = file(var.public_key)
}

resource "aws_instance" "myapp-server" {
  ami = "ami-05524d6658fcf35b6"
  instance_type = var.instance_type
  subnet_id = aws_subnet.myapp_subnet_1.id
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name = aws_key_pair.ssh-key.key_name
  user_data = file("./entry-script.sh")

  tags = { Name = "${var.env_prefix}-ec2-instance" }
}
```

- task1_entry_script.png

```
GNU nano 7.2 entry-script.sh *
#!/bin/bash
set -e
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx
```

```
@23-22411-057-sudo → ~/Lab12 $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:3qu6y+599xY9NCH3tgWSwdFe9M71Ggw1bJoZCzZiC0U codespace@codespaces-a377e3
The key's randomart image is:
+--[ED25519 256]--+
|      oE .o*o..|
| . o + =o=o+o|
| o + o.Xoo=|
| . =o.=*|
| S +oB|
| . . .|=|
| . . .|
| . . .|
| oB+oo.. o.|
+-----[SHA256]-----+
@23-22411-057-sudo → ~/Lab12 $
```

- task1_ssh_keygen.png

```
@23-22411-057-sudo → ~/Lab12 $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:3qu6y+599xY9NCH3tgWSwdFe9M71Ggw1bJoZCzZiC0U codespace@codespaces-a377e3
The key's randomart image is:
+--[ED25519 256]--+
|      oE  .o*o..|
|      . o + =o=+o|
|      o + o.Xoo=|
|      .  =o.=*|
|      S      +oB|
|      . .  .=.|
|      . .  ...|
|      . .  ...|
|      oB+oo.. o.|
+-----[SHA256]-----+
```

- task1_terraform_init.png

```
@23-22411-057-sudo → ~/Lab12 $ terraform init

Initializing the backend...

Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Finding latest version of hashicorp/http...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo → ~/Lab12 $ |
```

- task1_terraform_apply.png


```
@23-22411-057-sudo → ~/Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
+   arn                = (known after apply)
+   default_route_table_id = (known after apply)
+   id                 = (known after apply)
+   owner_id           = (known after apply)
+   region              = "me-central-1"
+   route               = [
+     {
+       cidr_block          = "0.0.0.0/0"
+       core_network_arn    = ""
+       destination_prefix_list_id = ""
+       egress_only_gateway_id = ""
+       gateway_id          = (known after apply)
+       instance_id         = ""
+       ipv6_cidr_block      = ""
+       nat_gateway_id      = ""
+       network_interface_id = ""
+       transit_gateway_id   = ""
+       vpc_endpoint_id     = ""
+       vpc_peering_connection_id = ""
+     },
+   ]
+   tags                = {
+     "Name" = "dev-rt"
+   }
+   tags_all            = {
+     "Name" = "dev-rt"
+   }
+   vpc_id              = (known after apply)
}

# aws_default_security_group.default_sg will be created
+ resource "aws_default_security_group" "default_sg" {
+   arn                = (known after apply)
+   description         = (known after apply)
+   egress              = [
+     {
+       cidr_blocks        = [
+         "0.0.0.0/0",
+       ]
+       description         = ""
+       from_port           = 0
+       ipv6_cidr_blocks    = []
+       prefix_list_ids     = []
+       protocol            = "-1"
+       security_groups     = []
+     },
+   ]
}
```

```

+ arn = (known after apply)
+ cidr_block = "10.0.0.0/16"
+ default_network_acl_id = (known after apply)
+ default_route_table_id = (known after apply)
+ default_security_group_id = (known after apply)
+ dhcp_options_id = (known after apply)
+ enable_dns_hostnames = (known after apply)
+ enable_dns_support = true
+ enable_network_address_usage_metrics = (known after apply)
+ id = (known after apply)
+ instance_tenancy = "default"
+ ipv6_association_id = (known after apply)
+ ipv6_cidr_block = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id = (known after apply)
+ owner_id = (known after apply)
+ region = "me-central-1"
+ tags = {
+   + "Name" = "dev-vpc"
+ }
+ tags_all = {
+   + "Name" = "dev-vpc"
+ }
}

```

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ aws_instance_public_ip = (known after apply)
aws_key_pair.ssh-key: Creating...
aws_vpc.myapp_vpc: Creating...
aws_key_pair.ssh-key: Creation complete after 1s [id=serverkey]
aws_vpc.myapp_vpc: Creation complete after 1s [id=vpc-0edbf5c7fcf64840f]
aws_internet_gateway.myapp_igw: Creating...
aws_subnet.myapp_subnet_1: Creating...
aws_default_security_group.default_sg: Creating...
aws_subnet.myapp_subnet_1: Creation complete after 1s [id=subnet-08f9f6834608ccc59]
aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-0c4bc43a4e90586f8]
aws_default_route_table.main_rt: Creating...
aws_default_route_table.main_rt: Creation complete after 2s [id=rtb-0d48439a0e3b5d47e]
aws_default_security_group.default_sg: Creation complete after 3s [id=sg-0fbfedbc153287480]
aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still creating... [10s elapsed]
aws_instance.myapp-server: Creation complete after 13s [id=i-0d09eaaaf26f80d22b]

```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

```

aws_instance_public_ip = "3.28.46.20"
@23-22411-057-sudo → ~/Lab12 $

```

- task1_terraform_output.png

```

@23-22411-057-sudo → ~/Lab12 $ terraform output
aws_instance_public_ip = "3.28.46.20"
@23-22411-057-sudo → ~/Lab12 $

```

- task1_nginx_browser.png



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

- task1_terraform_destroy.png

```

@23-22411-057-sudo → ~/Lab12 $ terraform destroy
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
aws_key_pair.ssh-key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0edbf5c7fcf64840f]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0c4bc43a4e90586f8]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-08f9f6834608ccc59]
aws_default_security_group.default_sg: Refreshing state... [id=sg-0fbfedbc153287480]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d48439a0e3b5d47e]
aws_instance.myapp-server: Refreshing state... [id=i-0d09eaaf26f80d22b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn                = "arn:aws:ec2:me-central-1:359416636630:route-table/rtb-0d48439a0e3b5d47e" -> null
  - default_route_table_id = "rtb-0d48439a0e3b5d47e" -> null
  - id                  = "rtb-0d48439a0e3b5d47e" -> null
  - owner_id            = "359416636630" -> null
  - propagating_vgws     = [] -> null
  - region              = "me-central-1" -> null
  - route               = [
    - {
      - cidr_block          = "0.0.0.0/0"
      - core_network_arn    = ""
      - destination_prefix_list_id = ""
      - egress_only_gateway_id = ""
      - gateway_id          = "igw-0c4bc43a4e90586f8"
      - instance_id         = ""
      - ipv6_cidr_block      = ""
      - nat_gateway_id      = ""
      - network_interface_id = ""
      - transit_gateway_id   = ""
      - vpc_endpoint_id      = ""
      - vpc_peering_connection_id = ""
    }
  ] -> null
  - tags                = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all            = {
    - "Name" = "dev-rt"
  }
}

```

Enter a value: yes

```

aws_default_route_table.main_rt: Destroying... [id=rtb-0d48439a0e3b5d47e]
aws_instance.myapp-server: Destroying... [id=i-0d09eaaf26f80d22b]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-0c4bc43a4e90586f8]
aws_instance.myapp-server: Still destroying... [id=i-0d09eaaf26f80d22b, 10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0c4bc43a4e90586f8, 10s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-0d09eaaf26f80d22b, 20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0c4bc43a4e90586f8, 20s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 28s
aws_instance.myapp-server: Still destroying... [id=i-0d09eaaf26f80d22b, 30s elapsed]
aws_instance.myapp-server: Destruction complete after 31s
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-08f9f6834608ccc59]
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_default_security_group.default_sg: Destroying... [id=sg-0fbfedbc153287480]
aws_default_security_group.default_sg: Destruction complete after 0s
aws_key_pair.ssh-key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0edbf5c7fcf64840f]
aws_vpc.myapp_vpc: Destruction complete after 1s

```

Destroy complete! Resources: 7 destroyed.

```
@23-22411-057-sudo → ~/Lab12 $ |
```

Task 2 — Use remote-exec provisioner

task2_main_tf_remote_exec.png

```

GNU nano 7.2                                     main.tf *
}

resource "aws_key_pair" "ssh-key" {
  key_name   = "serverkey"
  public_key = file(var.public_key)
}

resource "aws_instance" "myapp-server" {
  ami                = "ami-05524d6658fcf35b6"
  instance_type      = var.instance_type
  subnet_id          = aws_subnet.myapp_subnet_1.id
  security_groups    = [aws_default_security_group.default_sg.id]
  availability_zone  = var.availability_zone
  associate_public_ip_address = true
  key_name           = aws_key_pair.ssh-key.key_name

  connection {
    type     = "ssh"
    user     = "ec2-user"
    private_key = file(var.private_key)
    host     = self.public_ip
  }

  provisioner "remote-exec" {
    inline = [
      "sudo yum update -y",
      "sudo yum install -y nginx",
      "sudo systemctl start nginx",
      "sudo systemctl enable nginx"
    ]
  }

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

task2_terraform_apply.png

```

@23-22411-057-sudo → ~/Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
  + arn                = (known after apply)
  + default_route_table_id = (known after apply)
  + id                 = (known after apply)
  + owner_id           = (known after apply)
  + region             = "me-central-1"
  + route              = [
    + {
      + cidr_block          = "0.0.0.0/0"
      + core_network_arn    = ""
      + destination_prefix_list_id = ""
      + egress_only_gateway_id = ""
      + gateway_id          = (known after apply)
      + instance_id         = ""
      + ipv6_cidr_block      = ""
      + nat_gateway_id      = ""
      + network_interface_id = ""
      + transit_gateway_id   = ""
      + vpc_endpoint_id     = ""
      + vpc_peering_connection_id = ""
    }
  ],
  + tags               = {
    + "Name" = "dev-rt"
  }
  + tags_all           = {
    + "Name" = "dev-rt"
  }
  + vpc_id              = (known after apply)
}

# aws_default_security_group.default_sg will be created
+ resource "aws_default_security_group" "default_sg" {
  + arn                = (known after apply)
  + description         = (known after apply)
  + egress              = [
    + {
      + cidr_blocks = [
        + "0.0.0.0/0",
      ]
    }
  ]
}

```

task2_terraform_output.png


```
aws_instance.myapp-server (remote-exec): Installed:
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
aws_instance.myapp-server: Creation complete after 31s [id=i-02a948251d7fad319]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:
aws_instance_public_ip = "3.28.205.138"
@23-22411-057-sudo → ~/Lab12 $ terraform output
aws_instance_public_ip = "3.28.205.138"
@23-22411-057-sudo → ~/Lab12 $ |
```

task2_nginx_browser.png



Task 3 — Use file and local-exec provisioners

task3_main_tf_all_provisioners.png

```
GNU nano 7.2 main.tf
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block

  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id            = aws_vpc.myapp_vpc.id
  cidr_block        = var.subnet_cidr_block
  availability_zone = var.availability_zone

  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = aws_vpc.myapp_vpc.id

  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_default_route_table" "main_rt" {
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

resource "aws_default_security_group" "default_sg" {
  vpc_id = aws_vpc.myapp_vpc.id
}
```

```

GNU nano 7.2                                                                    main.tf *
}

resource "aws_key_pair" "ssh-key" {
  key_name   = "serverkey"
  public_key = file(var.public_key)
}

resource "aws_instance" "myapp-server" {
  ami               = "ami-05524d6658fcf35b6"
  instance_type     = var.instance_type
  subnet_id         = aws_subnet.myapp_subnet_1.id
  security_groups   = [aws_default_security_group.default_sg.id]
  availability_zone  = var.availability_zone
  associate_public_ip_address = true
  key_name          = aws_key_pair.ssh-key.key_name

  connection {
    type     = "ssh"
    user     = "ec2-user"
    private_key = file(var.private_key)
    host      = self.public_ip
  }

  provisioner "file" {
    source     = "./entry-script.sh"
    destination = "/home/ec2-user/entry-script-on-ec2.sh"
  }

  provisioner "remote-exec" {
    inline = [
      "sudo chmod +x /home/ec2-user/entry-script-on-ec2.sh",
      "sudo /home/ec2-user/entry-script-on-ec2.sh"
    ]
  }

  provisioner "local-exec" {
    command = <<-EOF
    echo Instance ${self.id} with public IP ${self.public_ip} has been created
    EOF
  }

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

task3_terraform_apply.png

```

023-22411-057~sudo → ~/Lab12 $ terraform init

Initializing the backend...

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
023-22411-057~sudo → ~/Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
aws_key_pair.ssh-key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-09b385d6a000bc6d9]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-042499f36f8318127]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0d935172076d4eec0]
aws_default_security_group.default_sg: Refreshing state... [id=sg-09cd20a9d6604c00e]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d4e0193a02cd2677]
aws_instance.myapp-server: Refreshing state... [id=i-02a948251d7fad319]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

# aws_instance.myapp-server must be replaced
-/+ resource "aws_instance" "myapp-server" {
  ~ ann               = "arn:aws:ec2:me-central-1:359416636630:instance/i-02a948251d7fad319" -> (known after apply)
  ~ disable_api_stop  = false -> (known after apply)
  ~ disable_api_termination = false -> (known after apply)
  ~ ebs_optimized     = false -> (known after apply)
  + enable_primary_ipv6 = (known after apply)
  ~ hibernation       = false -> null
  + host_id           = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile = (known after apply)
  ~ id                = "i-02a948251d7fad319" -> (known after apply)
  ~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)
  + instance_lifecycle = (known after apply)
  ~ instance_state     = "running" -> (known after apply)
}

```

task3_terraform_output.png

```
aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
aws_instance.myapp-server: Provisioning with 'local-exec'...
aws_instance.myapp-server (local-exec): Executing: ["/bin/sh" "-c" "echo Instance i-020b899480cc9da20 with public IP 3.29.21.183 has been created\n"]
aws_instance.myapp-server (local-exec): Instance i-020b899480cc9da20 with public IP 3.29.21.183 has been created
aws_instance.myapp-server: Creation complete after 33s [id=i-020b899480cc9da20]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

Outputs:

aws_instance_public_ip = "3.29.21.183"
@23-22411-057-sudo → ~/Lab12 $ terraform output
aws_instance_public_ip = "3.29.21.183"
@23-22411-057-sudo → ~/Lab12 $
```

task3_nginx_browser.png



task3_terraform_destroy.png

```
@23-22411-057-sudo → ~/Lab12 $ terraform destroy
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
aws_key_pair.ssh-key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-09b385d6a000bc6d9]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0d935172076d4eec0]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-042499f36f8318127]
aws_default_security_group.default_sg: Refreshing state... [id=sg-09cd20a9d6604c00e]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d4e0193a02cd2677]
aws_instance.myapp-server: Refreshing state... [id=i-020b899480cc9da20]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:route-table/rtb-0d4e0193a02cd2677" -> null
  - default_route_table_id = "rtb-0d4e0193a02cd2677" -> null
  - id = "rtb-0d4e0193a02cd2677" -> null
  - owner_id = "359416636630" -> null
  - propagating_vgws = [] -> null
  - region = "me-central-1" -> null
  - route = [
    - {
      - cidr_block = "0.0.0.0/0"
      - core_network_arn = ""
      - destination_prefix_list_id = ""
      - egress_only_gateway_id = ""
      - gateway_id = "igw-042499f36f8318127"
      - instance_id = ""
      - ipv6_cidr_block = ""
      - nat_gateway_id = ""
      - network_interface_id = ""
      - transit_gateway_id = ""
      - vpc_endpoint_id = ""
      - vpc_peering_connection_id = ""
    }
  ] -> null
  - tags = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all = {
    - "Name" = "dev-rt"
  } -> null
  - vpc_id = "vpc-09b385d6a000bc6d9" -> null
}

# aws_default_security_group.default_sg will be destroyed
```

task3_main_tf_restored.png


```

GNU nano 7.2 main.tf *
}

resource "aws_key_pair" "ssh-key" {
  key_name   = "serverkey"
  public_key = file(var.public_key)
}

resource "aws_instance" "myapp-server" {
  ami               = "ami-05524d6658fcf35b6"
  instance_type     = var.instance_type
  subnet_id         = aws_subnet.myapp_subnet_1.id
  security_groups   = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name          = aws_key_pair.ssh-key.key_name

  user_data = file("./entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

Task 4 — Create Terraform modules (subnet module)

- task4_module_structure.png

```

@23-22411-057-sudo → ~/Lab12 $ nano main.tf
@23-22411-057-sudo → ~/Lab12 $ mkdir -p modules/subnet
@23-22411-057-sudo → ~/Lab12 $ touch modules/subnet/main.tf
@23-22411-057-sudo → ~/Lab12 $ touch modules/subnet/variables.tf
@23-22411-057-sudo → ~/Lab12 $ tree modules
modules
├── subnet
│   ├── main.tf
│   └── variables.tf
2 directories, 2 files
@23-22411-057-sudo → ~/Lab12 $ touch modules/subnet/outputs.tf
@23-22411-057-sudo → ~/Lab12 $ tree modules
modules
├── subnet
│   ├── main.tf
│   ├── outputs.tf
│   └── variables.tf
2 directories, 3 files
@23-22411-057-sudo → ~/Lab12 $

```

- task4_subnet_variables.png

```

GNU nano 7.2 modules/subnet/variables.tf *
variable "vpc_id" {}

variable "subnet_cidr_block" {}

variable "availability_zone" {}

variable "env_prefix" {}

variable "default_route_table_id" {}

```

- task4_subnet_main.png

```

GNU nano 7.2                                     modules/subnet/main.tf *
resource "aws_subnet" "myapp_subnet_1" {
  vpc_id            = var.vpc_id
  cidr_block        = var.subnet_cidr_block
  availability_zone  = var.availability_zone
  map_public_ip_on_launch = true

  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = var.vpc_id

  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_default_route_table" "main_rt" {
  default_route_table_id = var.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

```

- task4_subnet_outputs.png

```

GNU nano 7.2                                     modules/subnet/outputs.tf *
output "subnet" {
  value = aws_subnet.myapp_subnet_1
}

```

- task4_main_tf_with_module.png

```

GNU nano 7.2                                     main.tf *
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

# -----
# VPC
# -----
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block

  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

# -----
# SUBNET MODULE
# -----
module "myapp-subnet" {
  source          = "./modules/subnet"
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix       = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

# -----
# SECURITY GROUP
# -----
resource "aws_default_security_group" "default_sg" {
  vpc_id = aws_vpc.myapp_vpc.id

  ingress {
    from_port = 22
    to_port   = 22
  }
}

```

- task4_terraform_init.png

```

GNU nano 7.2 main.tf *
    protocol      = "-1"
    cidr_blocks    = ["0.0.0.0/0"]
    prefix_list_ids = []
}

tags = {
    Name = "${var.env_prefix}-default-sg"
}
}

# -----
# KEY PAIR
# -----
resource "aws_key_pair" "ssh-key" {
    key_name     = "serverkey"
    public_key   = file(var.public_key)
}

# -----
# EC2 INSTANCE
# -----
resource "aws_instance" "myapp-server" {
    ami                  = "ami-05524d6658fcf35b6"
    instance_type       = var.instance_type
    subnet_id           = module.myapp-subnet.subnet.id
    security_groups     = [aws_default_security_group.default_sg.id]
    availability_zone   = var.availability_zone
    associate_public_ip_address = true
    key_name            = aws_key_pair.ssh-key.key_name

    user_data = file("./entry-script.sh")

    tags = {
        Name = "${var.env_prefix}-ec2-instance"
    }
}

```

- task4_terraform_apply.png

```

@23-22411-057-sudo → ~/Lab12 $ terraform init

Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo → ~/Lab12 $ |

```

- task4_terraform_output.png

```
@23-22411-057-sudo → ~/Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_default_security_group.default_sg will be created
+ resource "aws_default_security_group" "default_sg" {
+   arn                = (known after apply)
+   description        = (known after apply)
+   egress              = [
+     {
+       + cidr_blocks = [
+         + "0.0.0.0/0",
+       ]
+       + description = ""
+       + from_port   = 0
+       + ipv6_cidr_blocks = []
+       + prefix_list_ids = []
+       + protocol     = "-1"
+       + security_groups = []
+       + self         = false
+       + to_port      = 0
+     },
+   ]
+   id                = (known after apply)
+   ingress            = [
+     {
+       + cidr_blocks = [
+         + "0.0.0.0/0",
+       ]
+       + description = ""
+       + from_port   = 80
+       + ipv6_cidr_blocks = []
+       + prefix_list_ids = []
+       + protocol     = "tcp"
+       + security_groups = []
+       + self         = false
+     },
+   ]
+ }
```

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ aws_instance_public_ip = (known after apply)
aws_key_pair.ssh-key: Creating...
aws_vpc.myapp_vpc: Creating...
aws_key_pair.ssh-key: Creation complete after 0s [id=serverkey]
aws_vpc.myapp_vpc: Creation complete after 1s [id=vpc-0910ef80c25430004]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
aws_default_security_group.default_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-05c27d1507dcf661b]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-0741d1a92239926db]
aws_default_security_group.default_sg: Creation complete after 3s [id=sg-0abb3cb35e7ebabc6]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-0b50e00a1dedc9e17]
aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still creating... [10s elapsed]
aws_instance.myapp-server: Creation complete after 13s [id=i-05edc000780daeb82]
```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

```
aws_instance_public_ip = "3.28.132.108"
@23-22411-057-sudo → ~/Lab12 $ terraform output
aws_instance_public_ip = "3.28.132.108"
@23-22411-057-sudo → ~/Lab12 $
```

● task4_nginx_browser.png



Task 5 — Create webserver module

- task5_webserver_module_structure.png

```
@23-22411-057-sudo → ~/Lab12 $ tree modules
modules
├── subnet
│   ├── main.tf
│   ├── outputs.tf
│   └── variables.tf
└── 2 directories, 3 files
@23-22411-057-sudo → ~/Lab12 $
```

```
@23-22411-057-sudo → ~/Lab12 $ nano modules/webserver/variables.tf
@23-22411-057-sudo → ~/Lab12 $ mkdir -p ~/Lab12/modules/webserver
@23-22411-057-sudo → ~/Lab12 $ cd ~/Lab12/modules/webserver
@23-22411-057-sudo → ~/Lab12/modules/webserver $ touch main.tf variables.tf outputs.tf
@23-22411-057-sudo → ~/Lab12/modules/webserver $
```

- task5_webserver_variables.png

```
GNU nano 7.2 variables.tf *
variable "env_prefix" {}
variable "instance_type" {}
variable "availability_zone" {}
variable "public_key" {}
variable "my_ip" {}
variable "vpc_id" {}
variable "subnet_id" {}
variable "script_path" {}
variable "instance_suffix" {}
|
```

- task5_webserver_main.png

```

GNU nano 7.2                                                                    main.tf
resource "aws_security_group" "web_sg" {
  vpc_id      = var.vpc_id
  name        = "${var.env_prefix}-web-sg-${var.instance_suffix}"
  description = "Security group for web server allowing HTTP, HTTPS and SSH"

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = [var.my_ip]
  }

  ingress {
    from_port = 443
    to_port   = 443
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
    prefix_list_ids = []
  }

  tags = {
    Name = "${var.env_prefix}-default-sg"
  }
}

```

- task5_webserver_outputs.png

```

GNU nano 7.2                                                                    outputs.tf *
output "aws_instance" {
  value = aws_instance.myapp-server
}
|

```

- task5_main_tf_webserver_module.png

```

GNU nano 7.2                                     main.tf *
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source          = "./modules/subnet"
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix       = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source          = "./modules/webserver"
  env_prefix       = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "./entry-script.sh"
  instance_suffix = ""
}

```

- task5_outputs_updated.png

```

GNU nano 7.2                                     outputs.tf *
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}

```

- task5_terraform_init.png

```

@23-22411-057-sudo → ~/Lab12 $ terraform init

Initializing the backend...
Initializing modules...

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo → ~/Lab12 $ |

```

- task5_terraform_apply.png

```
@23-22411-057-sudo → ~/Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
aws_key_pair.ssh-key: Refreshing state... [id=serverkey]
aws_default_security_group.default_sg: Refreshing state... [id=sg-0abb3cb35e7ebabc6]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0910ef80c25430004]
aws_instance.myapp-server: Refreshing state... [id=i-05edc000780daeb82]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0b50e00a1dedc9e17]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-05c27d1507dcf661b]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0741d1a92239926db]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
- destroy

Terraform will perform the following actions:

# aws_default_security_group.default_sg will be destroyed
# (because aws_default_security_group.default_sg is not in configuration)
- resource "aws_default_security_group" "default_sg" {
  - arn              = "arn:aws:ec2:me-central-1:359416636630:security-group/sg-0abb3cb35e7ebabc6" -> null
  - description      = "default VPC security group" -> null
  - egress           = [
    - {
      - cidr_blocks    = [
        - "0.0.0.0/0",
      ]
      - description    = ""
      - from_port      = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids = []
      - protocol       = "-1"
      - security_groups = []
      - self           = false
      - to_port        = 0
    }
  ],
  -> null
- id              = "sg-0abb3cb35e7ebabc6" -> null
- ingress         = [
  - {
    - cidr_blocks    = [
      - "0.0.0.0/0",
    ]
    - description    = ""
    - from_port      = 80
  }
]
```

- task5_terraform_output.png

```
Changes to Outputs:
- aws_instance_public_ip = "3.28.132.108" -> null
+ webserver_public_ip    = (known after apply)
module.myapp-webserver.aws_key_pair.ssh-key: Creating...
aws_instance.myapp-server: Destroying... [id=i-05edc000780daeb82]
module.myapp-webserver.aws_security_group.web_sg: Creating...
module.myapp-webserver.aws_key_pair.ssh-key: Creation complete after 1s [id=dev-serverkey-0]
module.myapp-webserver.aws_security_group.web_sg: Creation complete after 5s [id=sg-013c78331d8880eb3]
module.myapp-webserver.aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still destroying... [id=i-05edc000780daeb82, 10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still creating... [10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Creation complete after 13s [id=i-091a3cba1b026ae9c]
aws_instance.myapp-server: Still destroying... [id=i-05edc000780daeb82, 20s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-05edc000780daeb82, 30s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-05edc000780daeb82, 40s elapsed]
aws_instance.myapp-server: Destruction complete after 41s
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_default_security_group.default_sg: Destroying... [id=sg-0abb3cb35e7ebabc6]
aws_default_security_group.default_sg: Destruction complete after 0s
aws_key_pair.ssh-key: Destruction complete after 0s

Apply complete! Resources: 3 added, 0 changed, 3 destroyed.

Outputs:

webserver_public_ip = "40.172.187.73"
@23-22411-057-sudo → ~/Lab12 $ terraform output
webserver_public_ip = "40.172.187.73"
@23-22411-057-sudo → ~/Lab12 $ |
```

- task5_nginx_browser.png



- task5_terraform_destroy.png

```
@23-22411-057-sudo → ~/Lab12 $ terraform destroy
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
module.myapp-webserver.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0910ef80c25430004]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-05c27d1507dcf661b]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0b50e00a1dedc9e17]
module.myapp-webserver.aws_security_group.web_sg: Refreshing state... [id=sg-013c78331d8880eb3]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0741d1a92239926db]
module.myapp-webserver.aws_instance.myapp-server: Refreshing state... [id=i-091a3cbalb026ae9c]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be destroyed
- resource "aws_vpc" "myapp_vpc" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-0910ef80c25430004" -> null
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-08216d466535c7e18" -> null
  - default_route_table_id = "rtb-0741d1a92239926db" -> null
  - default_security_group_id = "sg-0abb3cb35e7ebabc6" -> null
  - dhcp_options_id = "dopt-07a746a94dc2af666" -> null
  - enable_dns_hostnames = false -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id = "vpc-0910ef80c25430004" -> null
  - instance_tenancy = "default" -> null
  - ipv6_netmask_length = 0 -> null
  - main_route_table_id = "rtb-0741d1a92239926db" -> null
  - owner_id = "359416636630" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "dev-vpc"
  } -> null
  - tags_all = {
    - "Name" = "dev-vpc"
  } -> null
}

# module.myapp-subnet.aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:route-table/rtb-0741d1a92239926db" -> null
}
```

Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:

- webserver_public_ip = "40.172.187.73" -> null

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
module.myapp-subnet.aws_default_route_table.main_rt: Destroying... [id=rtb-0741d1a92239926db]
module.myapp-webserver.aws_instance.myapp-server: Destroying... [id=i-091a3cbalb026ae9c]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-05c27d1507dcf661b]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-091a3cbalb026ae9c, 10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-05c27d1507dcf661b, 10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-091a3cbalb026ae9c, 20s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-05c27d1507dcf661b, 20s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 29s
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-091a3cbalb026ae9c, 30s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-091a3cbalb026ae9c, 40s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Destruction complete after 41s
module.myapp-webserver.aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0b50e00a1dedc9e17]
module.myapp-webserver.aws_security_group.web_sg: Destroying... [id=sg-013c78331d8880eb3]
module.myapp-webserver.aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver.aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0910ef80c25430004]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@23-22411-057-sudo → ~/Lab12 $
```

Task 6 — Configure HTTPS with self-signed certificates

- task6_entry_script_https.png


```
GNU nano 7.2                                     entry-script.sh *
#!/bin/bash
set -e

# Update system and install nginx
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx

# Create directories for SSL certificates
mkdir -p /etc/ssl/private
mkdir -p /etc/ssl/certs

# Get IMDSv2 token for metadata
TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

# Get public IP and hostname
PUBLIC_IP=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/public-ipv4)

PUBLIC_HOSTNAME=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/public-hostname)

# Generate self-signed SSL certificate
openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/ssl/private/selfsigned.key \
-out /etc/ssl/certs/selfsigned.crt \
-subj "/CN=$PUBLIC_IP" \
-addext "subjectAltName=IP:$PUBLIC_IP" \
-addext "basicConstraints=CA:FALSE" \
-addext "keyUsage=digitalSignature,keyEncipherment" \
-addext "extendedKeyUsage=serverAuth"

echo "Self-signed certificate created for IP: $PUBLIC_IP"

# Backup existing nginx.conf
cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.bak

# Overwrite nginx.conf for HTTPS
cat <<EOF > /etc/nginx/nginx.conf
```

```

GNU nano 7.2                                     entry-script.sh *
worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
                    '$status $body_bytes_sent "$http_referer" '
                    '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    server {
        listen 443 ssl;
        server_name $PUBLIC_IP;
        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root /usr/share/nginx/html;
            index index.html;
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
EOF

# Restart Nginx to apply HTTPS config
systemctl restart nginx

```

- task6_terraform_apply.png

```

@23-22411-057-sudo → ~/Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be created
+ resource "aws_vpc" "myapp_vpc" {
  + arn                = (known after apply)
  + cidr_block         = "10.0.0.0/16"
  + default_network_acl_id = (known after apply)
  + default_route_table_id = (known after apply)
  + default_security_group_id = (known after apply)
  + dhcp_options_id     = (known after apply)
  + enable_dns_hostnames = (known after apply)
  + enable_dns_support   = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                  = (known after apply)
  + instance_tenancy    = "default"
  + ipv6_association_id = (known after apply)
  + ipv6_cidr_block      = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id = (known after apply)
  + owner_id             = (known after apply)
  + region               = "me-central-1"
  + tags                 = {
    + "Name" = "dev-vpc"
  }
  + tags_all              = {
    + "Name" = "dev-vpc"
  }
}

# module.myapp-subnet.aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
  + arn                = (known after apply)
  + default_route_table_id = (known after apply)
  + id                  = (known after apply)
  + owner_id             = (known after apply)
  + region               = "me-central-1"
  + route               = [
    + {

```

```

Changes to Outputs:
+ webserver_public_ip = (known after apply)
module.myapp-webserver.aws_key_pair.ssh-key: Creating...
aws_vpc.myapp_vpc: Creating...
module.myapp-webserver.aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Creation complete after 2s [id=vpc-07a099dce71ef997f]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-webserver.aws_security_group.web_sg: Creating...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-06cbb71eaa6dfc965]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 2s [id=rtb-074182dc4de0b7541]
module.myapp-webserver.aws_security_group.web_sg: Creation complete after 4s [id=sg-07c4c44a2e018931b]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 12s [id=subnet-00ad9420e7b88ffe5]
module.myapp-webserver.aws_instance.myapp-server: Creating...
module.myapp-webserver.aws_instance.myapp-server: Still creating... [10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Creation complete after 13s [id=i-0b8fb5bb66f2b014e]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:
webserver_public_ip = "51.112.48.244"
@23-22411-057-sudo → ~/Lab12 $

```

- task6_terraform_output.png

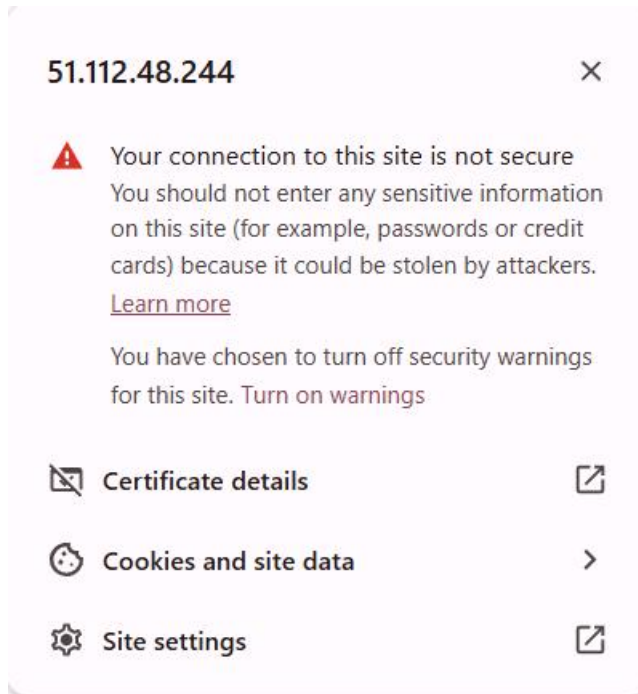
```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:
webserver_public_ip = "51.112.48.244"
@23-22411-057-sudo → ~/Lab12 $ terraform output
webserver_public_ip = "51.112.48.244"
@23-22411-057-sudo → ~/Lab12 $ |

```

- task6_browser_security_warning.png



- task6_nginx_https_browser.png

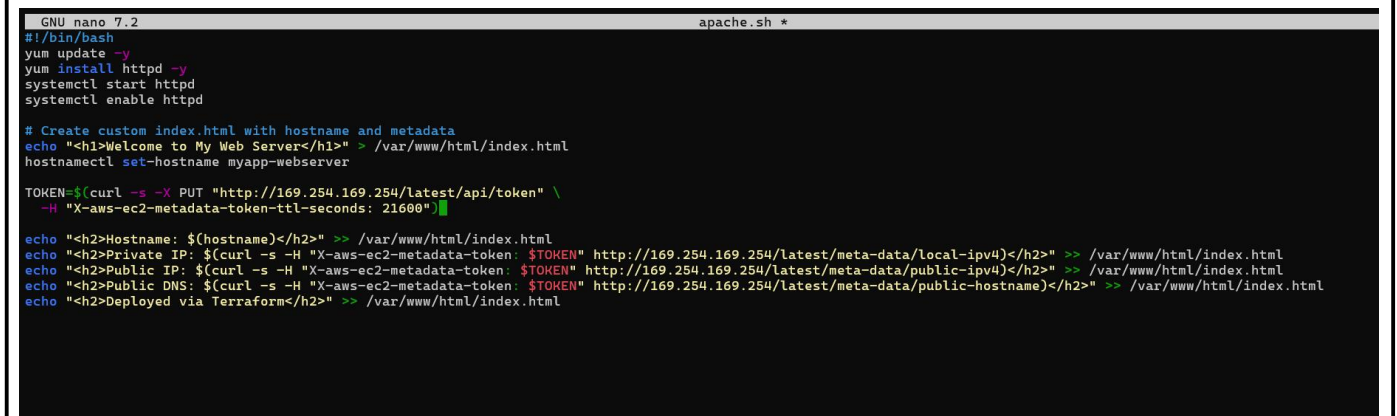


- task6_http_redirect.png



Task 7 — Configure Nginx as reverse proxy

- task7_apache_script.png



- task7_main_tf_web1.png

```

GNU nano 7.2                                     main.tf *
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source           = "./modules/subnet"
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix       = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source           = "./modules/webserver"
  env_prefix       = var.env_prefix
  instance_type    = var.instance_type
  availability_zone = var.availability_zone
  public_key       = var.public_key
  my_ip            = local.my_ip
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_id        = module.myapp-subnet.subnet.id
  script_path      = "./entry-script.sh"
  instance_suffix  = "0"
}

module "myapp-web-1" {
  source           = "./modules/webserver"
  env_prefix       = var.env_prefix
  instance_type    = var.instance_type
  availability_zone = var.availability_zone
  public_key       = var.public_key
  my_ip            = local.my_ip
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_id        = module.myapp-subnet.subnet.id
  script_path      = "./apache.sh"
  instance_suffix  = "1"
}

```

- task7_outputs_web1.png

```

GNU nano 7.2                                     outputs.tf *
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}

output "aws_web-1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}

```

- task7_terraform_apply.png


```

error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    server {
        listen 443 ssl;
        server_name 51.112.48.244;
        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root /usr/share/nginx/html;
            index index.html;
            proxy_pass http://3.29.124.127:80;
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
~
~
-- INSERT --

```

- task7_nginx_restart.png

```

[ec2-user@ip-10-0-10-50 ~]$ sudo vim /etc/nginx/nginx.conf
[ec2-user@ip-10-0-10-50 ~]$ sudo systemctl restart nginx
[ec2-user@ip-10-0-10-50 ~]$ |

```

- task7_error_log.png

```
[ec2-user@ip-10-0-10-50 ~]$ cat /var/log/nginx/error.log
2026/01/05 15:01:31 [notice] 2981#2981: using the "epoll" event method
2026/01/05 15:01:31 [notice] 2981#2981: nginx/1.28.0
2026/01/05 15:01:31 [notice] 2981#2981: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/05 15:01:31 [notice] 2981#2981: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/05 15:01:31 [notice] 3023#3023: start worker processes
2026/01/05 15:01:31 [notice] 3023#3023: start worker process 3025
2026/01/05 15:01:31 [notice] 3023#3023: start worker process 3026
2026/01/05 15:01:32 [notice] 3023#3023: signal 3 (SIGQUIT) received from 1, shutting down
2026/01/05 15:01:32 [notice] 3025#3025: gracefully shutting down
2026/01/05 15:01:32 [notice] 3025#3025: exiting
2026/01/05 15:01:32 [notice] 3025#3025: exit
2026/01/05 15:01:32 [notice] 3026#3026: gracefully shutting down
2026/01/05 15:01:32 [notice] 3026#3026: exiting
2026/01/05 15:01:32 [notice] 3026#3026: exit
2026/01/05 15:01:32 [notice] 3023#3023: signal 17 (SIGCHLD) received from 3025
2026/01/05 15:01:32 [notice] 3023#3023: worker process 3025 exited with code 0
2026/01/05 15:01:32 [notice] 3023#3023: signal 29 (SIGIO) received
2026/01/05 15:01:32 [notice] 3023#3023: signal 17 (SIGCHLD) received from 3026
2026/01/05 15:01:32 [notice] 3023#3023: worker process 3026 exited with code 0
2026/01/05 15:01:32 [notice] 3023#3023: exit
2026/01/05 15:01:32 [notice] 4078#4078: using the "epoll" event method
2026/01/05 15:01:32 [notice] 4078#4078: nginx/1.28.0
2026/01/05 15:01:32 [notice] 4078#4078: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/05 15:01:32 [notice] 4078#4078: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/05 15:01:32 [notice] 4119#4119: start worker processes
2026/01/05 15:01:32 [notice] 4119#4119: start worker process 4121
2026/01/05 15:01:32 [notice] 4119#4119: start worker process 4122
2026/01/05 15:02:38 [error] 4121#4121: *15 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 59.103.32.61, server: 51.112.48.244
2026/01/05 15:16:16 [notice] 4119#4119: signal 3 (SIGQUIT) received from 1, shutting down
2026/01/05 15:16:16 [notice] 4122#4122: gracefully shutting down
2026/01/05 15:16:16 [notice] 4122#4122: exiting
2026/01/05 15:16:16 [notice] 4122#4122: exit
2026/01/05 15:16:16 [notice] 4121#4121: gracefully shutting down
2026/01/05 15:16:16 [notice] 4121#4121: exiting
2026/01/05 15:16:16 [notice] 4121#4121: exit
2026/01/05 15:16:17 [notice] 4119#4119: signal 17 (SIGCHLD) received from 4122
2026/01/05 15:16:17 [notice] 4119#4119: worker process 4121 exited with code 0
2026/01/05 15:16:17 [notice] 4119#4119: worker process 4122 exited with code 0
2026/01/05 15:16:17 [notice] 4119#4119: exit
2026/01/05 15:16:17 [notice] 25693#25693: using the "epoll" event method
2026/01/05 15:16:17 [notice] 25693#25693: nginx/1.28.0
2026/01/05 15:16:17 [notice] 25693#25693: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/05 15:16:17 [notice] 25693#25693: getrlimit(RLIMIT_NOFILE): 65535:65535
```

- task7_access_log.png

```
[ec2-user@ip-10-0-10-50 ~]$ cat /var/log/nginx/access.log
59.103.32.61 - - [05/Jan/2026:15:02:28 +0000] "GET / HTTP/1.1" 301 169 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:02:36 +0000] "GET / HTTP/1.1" 200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:02:38 +0000] "GET / HTTP/1.1" 200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:02:38 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "https://51.112.48.244/"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:03:11 +0000] "GET / HTTP/1.1" 200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:03:23 +0000] "GET / HTTP/1.1" 200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:03:43 +0000] "GET / HTTP/1.1" 200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:05:32 +0000] "GET / HTTP/1.1" 200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
109.196.145.152 - - [05/Jan/2026:15:09:44 +0000] "GET / HTTP/1.0" 200 615 "-" "-" "-"
[ec2-user@ip-10-0-10-50 ~]$
```

- task7_mime_types.png

```
[ec2-user@ip-10-0-10-50 ~]$ cat /etc/nginx/mime.types
types {
application/A2L                                a2l;
application/AML                                aml;
application/andrew-inset                       ez;
application/ATF                                atf;
application/ATFX                               atfx;
application/ATXML                             atxml;
application/atom+xml                           atom;
application/atomcat+xml                       atomcat;
application/atomdeleted+xml                   atomdeleted;
application/atomsvc+xml                       atomsvc;
application/atsc-dwd+xml                      dwd;
application/atsc-held+xml                     held;
application/atsc-rsat+xml                     rsat;
application/auth-policy+xml                   apxml;
application/bacnet-xdd+zip                     xdd;
application/calendar+xml                      xcs;
application/cbor                               cbor;
application/cccx                               c3ex;
application/ccmp+xml                           ccmp;
application/ccxml+xml                         ccxml;
application/CDFX+XML                          cdfx;
application/cdmi-capability                   cdmia;
application/cdmi-container                    cdmic;
application/cdmi-domain                       cdmid;
application/cdmi-object                       cdmio;
application/cdmi-queue                        cdmiq;
application/CEA                               cea;
application/cellml+xml                        cellml cml;
application/clue_info+xml                     clue;
application/cms                               cmsc;
application/cpl+xml                           cpl;
application/csrattrs                          csrattrs;
application/dash+xml                          mpd;
application/dashdelta                         mpdd;
application/davmount+xml                      davmount;
application/DCD                               dcd;
application/dicom                             dcm;
application/dii                               dii;
application/DIT                               dit;
application/dskpp+xml                         xmls;
application/dssc+der                          dssc;
application/dssc+xml                          xdssc;
application/dvcs                             dvc;
```

- task7_ssl_cert.png

```
[ec2-user@ip-10-0-10-50 ~]$ cat /etc/ssl/certs/selfsigned.crt
-----BEGIN CERTIFICATE-----
MIIDPzCCAiegAwIBAgIUdXONuYVN3nF74W7K1/563/d0E4wDQYJKoZIhvcNAQEL
BQAwGDEWMBQGA1UEAwwNNTUuMTEyLjQ4LjI0NDAAeFw0yNjAxMDUxNTAxMzJaFw0y
NzAxMDUxNTAxMzJaMBGxGfjAUBGNVBAMMDUxLjExMi40OC4yNDQwggEiMA0GCSqG
SIb3DQEBAQUAA4IBDwAwggEKAoIBAQC0EReToiXnrG1C98gb3syZIB1QWBXmCAki
7GT1I2mBu5RxJ3kbj93Z2cIHXUgSUR34EUmTMU43IdEB1TCF94JNRoN542C+pWsm
ERgTtUFH7vdPpghmG+g/JyJ5Jww7p/14TX6pTyw9qdtUi+03PSXKAdsWZRIazdUF
/X9V2FTJhQFegz6LY8CJIV4pkbdoHLJDMMEjq4nYBVl99pBT1KJpoqThsv98LoPz
QMq7WcPw96WmTvn55jksms8acQsTJ5nCVbo3lqvoHd8K/XErsTqfTvOxMxG71zti
EmkL6TnBTqt4h58xnB5Aep6gRTyucZtteadApVRjyXooGLyxS9P7AgMBAAGjYAw
fjAdBgNVHQ4EFgQUZaQZioKVfuJkWTszSXP4GYVr9EAwHwYDVR0jBBgwFoAUZaQZ
ioKVfuJkWTszSXP4GYVr9EAwDwYDVR0RBAGwBocEM3Aw9DAJBGNVHRMEAjAAMAsG
A1UdDwQEAwIFoDATBgNVHUEDDAKBggrBgEFBQcDATANBgkqhkiG9w0BAQsFAAOC
AQEAgaUBuLGDsBWubXRcgn6NdrSnr/2KHBpBrR4zEjvtw2NELlya8yHja+8Iaifw
BvMMphCGoEtm7Ubq+/62ksKiKFSapDLqtMLgh+2xu9t07Rh3RxyifQr/HASJoeg9
beOEhoFvegdFMPPrmeowbe6gC+GAS68MVsvJ3h5geo67NjxSmULUzS5MFDzKZ0sj
yn7EICXZ3vKuB6PYGnjbdNoous2BZA9M8l+C2ZkQW65UVQy7yOY1t0+tllofcWFH
63I4MCZhERDPkfjag5bYEgq+WC43pjN7B5och0sJSEmQ/UlsvZa+w601dOnqoOvI
olrMUnRCgDZhdZ1+HFmaJogmQ==
-----END CERTIFICATE-----
[ec2-user@ip-10-0-10-50 ~]$ |
```

- task7_ssl_key.png


```
[ec2-user@ip-10-0-10-50 ~]$ sudo cat /etc/ssl/private/selfsigned.key
-----BEGIN PRIVATE KEY-----
MIIEvAIBADANBgkqhkiG9w0BAQEFAASCBAwggSiAgEAAoIBAQC0EReToiXnrG1C
98gb3syZIB1QWBXmCAki7GT1I2mBu5RxJ3kbj93Z2cIHxUgSUR34EUmTMU43IdEB
1TCF94JNRoN542C+pWsmERgTtUFH7vdPpghmG+g/JyJ5Jww7p/14TX6pTyw9qdtU
i+03PSXKAdsWZRIazdUF/X9V2FTJhQFegz6LY8CJIV4pkbdoHLJDMMEjq4nYBVl9
9pBT1KJpoqThsv98LoPzQM7WcPw96WmTvn5jksms8acQsTJ5nCvbo3lqvoHd8K
/XErsTqfTv0xMxG71ztiEmkL6TnBTqt4h58xnB5AEp6gRTyucZtteadApVRjyXoo
GLyxs9P7AgMBAAECggEAPBSz5CunMBzLPAIG/92gOWhOv58Ud3NY8RELQnH1Yzce
eX7tNtgS8vxodCKYyJvOPxYtkXJTIawj290xZBHIpVYQylumzRXMdmifuvuwc3r1
g9rnVIKdRG0fYIM53rHXrgtVZtKm9VA2gYhKtzKs6DEKU92XY+An4t7JXPBYogAy
w8yp09gz1NDu5VCiNNHTgMowLGDK+zaFSSu7kd+q5CnAcUkAKWXfyt+fZFk1uOAC
D2ZJhBzPX7ZKQqGyQ8IILAV/CHVLgY6v3aEeCG+pMXjVekDvhB21xWftD1UpoCOu
ZiAbJeDnSC4vfJgxEa3KBKac3Cti8FymDIrVOLuy2QKBgQDYW3JhUK6F5FUL/yNj
wu/jaIVoyWnZIBDqyXzDx+XnmILpmIZ+tjdudfnDm5LPqQ0DX8GZ079PgFTt1QF92
a88ZV53h/yyk0LUfZ1qUqyviK06WymiWqrMD3fTqRZem7aomSIJGwuBFwy9EDTKV
vLNTjn7MWLKV6Y91/YVvYW8MbQKBgQDQVD2Nzew9muI0DGsK3p23DzkC5UTzc3bpV
8NBndcekck/gPq3Wl/qm+p6LCGd/rq/4A/Gp637VFLKgKnNNJyMqK9znDcjetkFu
WLOGqKgSwa0pjzdlXQ5DfboGogwdloEfj2ecqXTIdM9ooymIJP00MkfK9qYrka5M
aKjkiixRBwKBgFE+tcR5YzAA0DeTDZU0xMpem4pn6AmS0wd3thRG4nx/dtmFXyWB
/PrPHFJ4t5ZF2qj8K11KGgQRtQwoqjBr948Kqc0N550bf9hVFVyykn4Q5NdWeBvQ
ZIK1PJ0kmPgkUzqhC9rKnzEsNqmVUyOLgs0jxNWzkrYh3K9wDkVvDyZZAoGABc5n
jWwbHVeGMEF4cPzVFvL1/Jbkk7i3LURkyY0CJc8fsiLDJgf6QratwB2chHREQ2QI
H/NU/4WeNtMxVTbt1GZhnMJZaQhafcFmCVNrUSeKConmnpKMovY4LZoY7m19k1ib
ID/V+OiGBnn71cP8CDsejb9vA9zyfQ6VG7VQQ90CgYADuVmQijMT7f8mt/dHEC1u
vk8Z0NGE9mnld1A5j7zxWKV+KXCduFGq1aWL980TMCDtL/hWYx/ISEiWLjFoPjFd
sU0gnVi8rMcFKHc8U+aGwmNePGzEfL/UNDm1csGsNkhA3960ooLR+7RKh9ybziGy
GzPpQJLMPgdptk90FyK5Lw==
-----END PRIVATE KEY-----
[ec2-user@ip-10-0-10-50 ~]$ |
```

- task7_reverse_proxy_browser.png



Task 8 — Configure Nginx as load balancer

- task8_main_tf_web2.png


```

GNU nano 7.2                                     main.tf *
env_prefix          = var.env_prefix
default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source          = "./modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "./entry-script.sh"
  instance_suffix = "0"
}

module "myapp-web-1" {
  source          = "./modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "./apache.sh"
  instance_suffix = "1"
}

module "myapp-web-2" {
  source          = "./modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "./apache.sh"
  instance_suffix = "2"
}

```

- task8_outputs_web2.png

```

GNU nano 7.2                                     outputs.tf *
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}

output "aws_web-1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}

output "aws_web-2_public_ip" {
  value = module.myapp-web-2.aws_instance.public_ip
}

```

- task8_terraform_apply.png

```

Initializing the backend...
Initializing modules...
- myapp-web-2 in modules/webserver

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/http v3.5.0

```

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
data.http.my_ip: Reading
data.http.my_ip: Read complete after 1s [id=https://icanhazip.com]
module.myapp-web-1.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-1]
module.myapp-webserver.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-07a099dce71ef997f]
module.myapp-subnet.aws_subnet.myapp_subnet-1: Refreshing state... [id=subnet-00ad9420e7b88ffe5]
module.myapp-web-1.aws_security_group.web_sg: Refreshing state... [id=sg-05f2e3ae936fd9c19]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-086cb71eaad6fc965]
module.myapp-webserver.aws_security_group.web_sg: Refreshing state... [id=sg-07c4uc44a2e018931b]
module.myapp-web-1.aws_instance.myapp-server: Refreshing state... [id=i-0a3568226f7fff15f]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-074182dc4de0b7541]
module.myapp-webserver.aws_instance.myapp-server: Refreshing state... [id=i-008f45bb6662b014e]
```

Terraform will perform the following actions:

- task8_terraform_output.png

```
aws_web-1_public_ip = "3.29.124.127"
aws_web-2_public_ip = "51.112.42.26"
webserver_public_ip = "51.112.48.244"
@23-22411-057-sudo → ~/Lab12 $ terraform output
aws_web-1_public_ip = "3.29.124.127"
aws_web-2_public_ip = "51.112.42.26"
webserver_public_ip = "51.112.48.244"
@23-22411-057-sudo → ~/Lab12 $
```

- task8 nginx conf load balancer.png

- task8 nginx restart.png

```
[ec2-user@ip-10-0-10-50 ~]$ sudo systemctl restart nginx
```

```

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include              /etc/nginx/mime.types;
    default_type         application/octet-stream;

    # Load balancing upstream block
    upstream backend_servers {
        server 3.29.124.127:80;
        server 51.112.42.26:80;
    }

    # HTTPS server block
    server {
        listen 443 ssl;
        server_name _;

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            proxy_pass http://backend_servers;
        }
    }
}
-- INSERT --

```

- task8_load_balancer_web1.png

```

[ec2-user@ip-10-0-10-50 ~]$ cat /var/log/nginx/access.log
59.103.32.61 - - [05/Jan/2026:15:02:28 +0000] "GET / HTTP/1.1" 301 169 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:02:36 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:02:38 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:02:38 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "https://51.112.48.244/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:03:13 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:03:21 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:03:43 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
59.103.32.61 - - [05/Jan/2026:15:05:32 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
109.196.145.152 - - [05/Jan/2026:15:09:44 +0000] "GET / HTTP/1.0" 200 615 "-" "-" "-"
87.236.176.158 - - [05/Jan/2026:15:37:34 +0000] "GET / HTTP/1.1" 301 169 "-" "Mozilla/5.0 (compatible; InternetMeasurement/1.0; +https://internet-measurement.com/)" "-"
87.236.176.158 - - [05/Jan/2026:15:37:35 +0000] "GET / HTTP/1.1" 200 189 "http://51.112.48.244/" "Mozilla/5.0 (compatible; InternetMeasurement/1.0; +https://internet-measurement.com/)" "-"
185.247.137.253 - - [05/Jan/2026:15:37:36 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "-" "Mozilla/5.0 (compatible; InternetMeasurement/1.0; +https://internet-measurement.com/)" "-"
45.55.186.92 - - [05/Jan/2026:15:37:37 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "-" "Mozilla/5.0 (compatible; InternetMeasurement/1.0; +https://internet-measurement.com/)" "-"

```

- task8_load_balancer_web2.png

```
[ec2-user@ip-10-0-10-50 ~]$ cat /var/log/nginx/error.log
2026/01/05 15:01:31 [notice] 2981#2981: using the "epoll" event method
2026/01/05 15:01:31 [notice] 2981#2981: nginx/1.28.0
2026/01/05 15:01:31 [notice] 2981#2981: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/05 15:01:31 [notice] 2981#2981: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/05 15:01:31 [notice] 3023#3023: start worker processes
2026/01/05 15:01:31 [notice] 3023#3023: start worker process 3025
2026/01/05 15:01:31 [notice] 3023#3023: start worker process 3026
2026/01/05 15:01:32 [notice] 3023#3023: signal 3 (SIGQUIT) received from 1, shutting down
2026/01/05 15:01:32 [notice] 3025#3025: gracefully shutting down
2026/01/05 15:01:32 [notice] 3025#3025: exiting
2026/01/05 15:01:32 [notice] 3025#3025: exit
2026/01/05 15:01:32 [notice] 3026#3026: gracefully shutting down
2026/01/05 15:01:32 [notice] 3026#3026: exiting
2026/01/05 15:01:32 [notice] 3026#3026: exit
2026/01/05 15:01:32 [notice] 3023#3023: signal 17 (SIGCHLD) received from 3025
2026/01/05 15:01:32 [notice] 3023#3023: worker process 3025 exited with code 0
2026/01/05 15:01:32 [notice] 3023#3023: signal 29 (SIGIO) received
2026/01/05 15:01:32 [notice] 3023#3023: signal 17 (SIGCHLD) received from 3026
2026/01/05 15:01:32 [notice] 3023#3023: worker process 3026 exited with code 0
2026/01/05 15:01:32 [notice] 3023#3023: exit
2026/01/05 15:01:32 [notice] 4078#4078: using the "epoll" event method
2026/01/05 15:01:32 [notice] 4078#4078: nginx/1.28.0
2026/01/05 15:01:32 [notice] 4078#4078: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/05 15:01:32 [notice] 4078#4078: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/05 15:01:32 [notice] 4119#4119: start worker processes
2026/01/05 15:01:32 [notice] 4119#4119: start worker process 4121
2026/01/05 15:01:32 [notice] 4119#4119: start worker process 4122
2026/01/05 15:02:38 [error] 4121#4121: *15 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 59.103.32.61, server: 51.112.48.244, request: "GET /favicon.ico HTTP/1.1", host: "51.112.48.244", referer: "https://51.112.48.244/"
2026/01/05 15:16:16 [notice] 4119#4119: signal 3 (SIGQUIT) received from 1, shutting down
2026/01/05 15:16:16 [notice] 4122#4122: gracefully shutting down
2026/01/05 15:16:16 [notice] 4122#4122: exiting
2026/01/05 15:16:16 [notice] 4122#4122: exit
2026/01/05 15:16:16 [notice] 4121#4121: gracefully shutting down
2026/01/05 15:16:16 [notice] 4121#4121: exiting
2026/01/05 15:16:16 [notice] 4121#4121: exit
2026/01/05 15:16:17 [notice] 4119#4119: signal 17 (SIGCHLD) received from 4122
2026/01/05 15:16:17 [notice] 4119#4119: worker process 4121 exited with code 0
2026/01/05 15:16:17 [notice] 4119#4119: worker process 4122 exited with code 0
2026/01/05 15:16:17 [notice] 4119#4119: exit
```

Task 9 — Configure high availability with backup servers

- task9_nginx_conf_ha_web1_primary.png

```
user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # High availability upstream: web-1 primary, web-2 backup
    upstream backend_servers {
        server 3.29.124.127:80; # web-1 primary
        server 51.112.42.26:80 backup; # web-2 backup
    }

    # HTTPS server block
    server {
        listen 443 ssl;
        server_name _;

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            proxy_pass http://backend_servers;
        }
    }
}
```

- task9_ha_web1_only.png

```
[ec2-user@ip-10-0-10-50 ~]$ sudo systemctl restart nginx
[ec2-user@ip-10-0-10-50 ~]$
```


Welcome to My Web Server

Hostname: myapp-webserver

Private IP: 10.0.10.187

Public IP: 3.29.124.127

Public DNS:

Deployed via Terraform

- task9_nginx_conf_ha_web2_primary.png

```
| user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # High availability upstream: web-2 primary, web-1 backup
    upstream backend_servers {
        server 3.29.124.127:80 backup; # web-1 backup
        server 51.112.42.26:80; # web-2 primary
    }

    # HTTPS server block
    server {
        listen 443 ssl;
        server_name _;

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            proxy_pass http://backend_servers;
        }
    }
}
-- INSERT --
```

- task9_ha_web2_only.png

```
[ec2-user@ip-10-0-10-50 ~]$ sudo vim /etc/nginx/nginx.conf
[ec2-user@ip-10-0-10-50 ~]$ sudo systemctl restart nginx
[ec2-user@ip-10-0-10-50 ~]$
```


Welcome to My Web Server

Hostname: myapp-webserver

Private IP: 10.0.10.187

Public IP: 3.29.124.127

Public DNS:

Deployed via Terraform

Task 10 — Enable Nginx caching

- task10_nginx_conf_cache.png

```
user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    # ----- CACHE CONFIGURATION -----
    proxy_cache_path /var/cache/nginx
        levels=1:2
        keys_zone=my_cache:10m
        inactive=60m
        max_size=1g;

    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # ----- BACKEND SERVERS -----
    upstream backend_servers {
        server 3.29.124.127:80;
        server 51.112.42.26:80;
    }

    # ----- HTTPS SERVER -----
    server {
```

- task10_nginx_restart.png

```
#
# ##
# #####
# #####
# #####
# #
# V
# --> https://aws.amazon.com/linux/amazon-linux-2023
```

```
Jan 05 16:25:33 ip-10-0-10-50.me-central-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jan 05 16:25:33 ip-10-0-10-50.me-central-1.compute.internal nginx[27857]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jan 05 16:25:33 ip-10-0-10-50.me-central-1.compute.internal nginx[27857]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jan 05 16:25:33 ip-10-0-10-50.me-central-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-10-0-10-50 ~]$
```

- task10_cache_miss.png

The screenshot shows a web browser window with the address bar displaying `https://51.112.48.244`. The page content includes:

- Welcome to My Web Server**
- Hostname: myapp-webserver**
- Private IP: 10.0.10.187**
- Public IP: 3.29.124.127**
- Public DNS:**
- Deployed via Terraform**

The Chrome DevTools Network tab is open, showing a list of requests. The first request, `51.112.48.244`, is selected, and its response is visible in the right pane. The response is an HTML document with a status of `200` and a content type of `text/html; charset=UTF-8`. The response body contains the same text as the page content shown in the browser window.

- task10_cache_hit.png

← → ↻ Not secure https://51.112.48.244

Welcome to My Web Server

Hostname: myapp-webserver

Private IP: 10.0.10.187

Public IP: 3.29.124.127

Public DNS:

Deployed via Terraform

Select a folder to store override files in Select folder

Elements Console Sources Network Performance >> 1

Filter

All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other

500 ms 1,000 ms 1,500 ms 2,000 ms 2,500 ms 3,000 ms

Name	Headers	Preview	Response	Initiator	Timing
51.112.48.244	Accept-Ranges: bytes Connection: keep-alive Content-Length: 189 Content-Type: text/html; charset=UTF-8 Date: Mon, 05 Jan 2026 16:28:56 GMT Etag: "bd-647a57a939177" Last-Modified: Mon, 05 Jan 2026 15:11:51 GMT Server: nginx/1.28.0 X-Cache-Status: HIT				

1 requests 467 B transferred 1 Request Headers Raw

Console AI assistance What's new Developer resources X

Filter by URL and error Load through website

Status	URL	Initiator	Total Bytes	Duration	Error
--------	-----	-----------	-------------	----------	-------

- task10_cache_directory.png

```
[ec2-user@ip-10-0-10-50 ~]$ sudo ls -la /var/cache/nginx/
total 0
drwx-----. 3 nginx root   15 Jan  5 16:26 .
drwxr-xr-x.  9 root  root  101 Jan  5 16:25 ..
drwx-----. 3 nginx nginx  16 Jan  5 16:26 4
[ec2-user@ip-10-0-10-50 ~]$ |
```

Cleanup

- cleanup_destroy_prompt.png


```

023-22411-057~sudo → ~/Lab12 $ terraform destroy
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
module.myapp-web-2.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-2]
module.myapp-webserver.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
module.myapp-web-1.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-1]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-07a099dce71ef997f]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-00ad9420e7b88ffe5]
module.myapp-web-2.aws_security_group.web_sg: Refreshing state... [id=sg-02fe26e03990b0949]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-06cbb71eaa6dfc965]
module.myapp-webserver.aws_security_group.web_sg: Refreshing state... [id=sg-07c4c44a2e018931b]
module.myapp-web-1.aws_security_group.web_sg: Refreshing state... [id=sg-05f2e3ae93efd9c19]
module.myapp-webserver.aws_instance.myapp-server: Refreshing state... [id=i-0b8fb5bb66f2b014e]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-074182dc4de0b7541]
module.myapp-web-2.aws_instance.myapp-server: Refreshing state... [id=i-098bdd1f1845539cc]
module.myapp-web-1.aws_instance.myapp-server: Refreshing state... [id=i-0a3568226ff7ff15f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be destroyed
- resource "aws_vpc" "myapp_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-07a099dce71ef997f" -> null
  - assign_generated_ipv6_cidr_block   = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id            = "acl-049f78186b7ad849c" -> null
  - default_route_table_id            = "rtb-074182dc4de0b7541" -> null
  - default_security_group_id         = "sg-039b86cc6bdcab04e" -> null
  - dhcp_options_id                  = "dopt-07a746a94dc2af666" -> null
  - enable_dns_hostnames              = false -> null
  - enable_dns_support                = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                                = "vpc-07a099dce71ef997f" -> null
  - instance_tenancy                  = "default" -> null
  - ipv6_netmask_length               = 0 -> null
  - main_route_table_id               = "rtb-074182dc4de0b7541" -> null
  - owner_id                          = "359416636630" -> null
  - region                            = "me-central-1" -> null
  - tags                              = {
    - "Name" = "dev-vpc"
  } -> null
  - tags_all                          = {
    - "Name" = "dev-vpc"
  } -> null
}

# module.myapp-subnet.aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {

```

- cleanup_destroy_complete.png

```

module.myapp-subnet.aws_default_route_table.main_rt: Destroying... [id=rtb-074182dc4de0b7541]
module.myapp-web-1.aws_instance.myapp-server: Destroying... [id=i-0a3568226ff7ff15f]
module.myapp-web-2.aws_instance.myapp-server: Destroying... [id=i-098bdd1f1845539cc]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-webserver.aws_instance.myapp-server: Destroying... [id=i-0b8fb5bb66f2b014e]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-06cbb71eaa6dfc965]
module.myapp-web-2.aws_instance.myapp-server: Still destroying... [id=i-098bdd1f1845539cc, 10s elapsed]
module.myapp-web-1.aws_instance.myapp-server: Still destroying... [id=i-0a3568226ff7ff15f, 10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-0b8fb5bb66f2b014e, 10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-06cbb71eaa6dfc965, 10s elapsed]
module.myapp-web-1.aws_instance.myapp-server: Still destroying... [id=i-0a3568226ff7ff15f, 20s elapsed]
module.myapp-web-2.aws_instance.myapp-server: Still destroying... [id=i-098bdd1f1845539cc, 20s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-0b8fb5bb66f2b014e, 20s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-06cbb71eaa6dfc965, 20s elapsed]
module.myapp-web-2.aws_instance.myapp-server: Still destroying... [id=i-098bdd1f1845539cc, 30s elapsed]
module.myapp-web-1.aws_instance.myapp-server: Still destroying... [id=i-0a3568226ff7ff15f, 30s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-0b8fb5bb66f2b014e, 30s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-06cbb71eaa6dfc965, 30s elapsed]
module.myapp-web-1.aws_instance.myapp-server: Still destroying... [id=i-0a3568226ff7ff15f, 40s elapsed]
module.myapp-web-2.aws_instance.myapp-server: Still destroying... [id=i-098bdd1f1845539cc, 40s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-0b8fb5bb66f2b014e, 40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-06cbb71eaa6dfc965, 40s elapsed]
module.myapp-web-2.aws_instance.myapp-server: Destruction complete after 40s
module.myapp-web-2.aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-2]
module.myapp-web-2.aws_security_group.web_sg: Destroying... [id=sg-02fe26e03990b0949]
module.myapp-web-2.aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-web-2.aws_security_group.web_sg: Destruction complete after 2s
module.myapp-web-1.aws_instance.myapp-server: Still destroying... [id=i-0a3568226ff7ff15f, 50s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still destroying... [id=i-0b8fb5bb66f2b014e, 50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-06cbb71eaa6dfc965, 50s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Destruction complete after 50s
module.myapp-webserver.aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver.aws_security_group.web_sg: Destroying... [id=sg-07c4c44a2e018931b]
module.myapp-webserver.aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-webserver.aws_security_group.web_sg: Destruction complete after 1s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 58s
module.myapp-web-1.aws_instance.myapp-server: Still destroying... [id=i-0a3568226ff7ff15f, 1m0s elapsed]
module.myapp-web-1.aws_instance.myapp-server: Destruction complete after 1m1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-00ad9420e7b88ffe5]
module.myapp-web-1.aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-1]
module.myapp-web-1.aws_security_group.web_sg: Destroying... [id=sg-05f2e3ae93efd9c19]
module.myapp-web-1.aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-web-1.aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-07a099dce71ef997f]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 13 destroyed.
023-22411-057~sudo → ~/Lab12 $ |

```

- cleanup_state_empty.png

```
@23-22411-057-sudo → ~/Lab12 $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.6.3",
  "serial": 95,
  "lineage": "766c71f6-f604-bb9b-a619-62a9f19ed74f",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@23-22411-057-sudo → ~/Lab12 $ |
```

- cleanup_final_files.png

```
@23-22411-057-sudo → ~/Lab12 $ ls -la
total 85992
drwxrwxr-x 5 codespace codespace 4096 Jan 5 16:40 .
drwxr-x--- 1 codespace codespace 4096 Jan 5 11:08 ..
drwxr-xr-x 4 codespace codespace 4096 Jan 5 12:43 .terraform
-rw-r--r-- 1 codespace codespace 2422 Jan 2 18:39 .terraform.lock.hcl
-rw-rw-r-- 1 codespace codespace 984 Jan 5 15:09 apache.sh
drwxr-xr-x 3 codespace codespace 4096 Dec 30 19:13 aws
-rw-rw-r-- 1 codespace codespace 63198381 Jan 2 18:47 awscliv2.zip
-rwxrwxr-x 1 codespace codespace 2294 Jan 5 15:00 entry-script.sh
-rw-rw-r-- 1 codespace codespace 123 Jan 2 18:34 locals.tf
-rw-rw-r-- 1 codespace codespace 1864 Jan 5 15:23 main.tf
drwxrwxr-x 4 codespace codespace 4096 Jan 5 14:35 modules
-rw-rw-r-- 1 codespace codespace 259 Jan 5 15:32 outputs.tf
-rw-rw-r-- 1 codespace codespace 175 Jan 5 15:54 outputs.tf.save
-rw-rw-r-- 1 codespace codespace 181 Jan 5 16:40 terraform.tfstate
-rw-rw-r-- 1 codespace codespace 40770 Jan 5 16:39 terraform.tfstate.backup
-rw-rw-r-- 1 codespace codespace 253 Jan 2 18:35 terraform.tfvars
-rw-rw-r-- 1 codespace codespace 24744282 Nov 1 2023 terraform_1.6.3_linux_amd64.zip
-rw-rw-r-- 1 codespace codespace 197 Jan 2 18:32 variables.tf
@23-22411-057-sudo → ~/Lab12 $
```